

The Normal distribution

Case applications

Example *Personnel tests*: Personnel tests are designed to test a job applicant's cognitive and/or physical abilities. An IQ test is an example of the former; a speed test involving the arrangement of pegs on a peg board is an example of the latter. During the 1990's, the proportion of employees using personnel tests dropped from more than 90% to less than 50%, in part due to concerns with equal rights laws. As a result of improved testing procedures and the realization that such test could hold down the costs associated with hiring the wrong person, the use of personnel tests increased dramatically presently.

A particular dexterity test is administered nationwide by a private testing service. It is known that for all tests administered last year the distribution of scores was approximately normal with mean 75 and a standard deviation 7.5

(a) A particular employer requires job candidates to score at least 80 on the dexterity test. Determine the percentage of the test scores during the past year that exceeded 80.

(b) The testing service reported to a particular employer that the score of one of its candidates fell at the 98th percentile of the distribution (that is, approximately 98% of the scores were lower than the candidate's, and only 2% were higher). Determine the candidate's score

Case: Rating employee performance: Almost all companies utilize some type of year-end performance review for their employees. Human Resource (HR) at an audit firm provides guidelines for supervisors rating their subordinates. For example, raters are advised to examine their ratings for a tendency to be either too lenient or too harsh. According to HR, “If you have this tendency, consider using a normal distribution; rate 10% of employees as exemplary, 20% distinguished, 40% competent, 20% marginal, and 10% unacceptable.” Suppose you are rating an employee’s performance on a scale of 1 (lowest) to 100 (highest). Also, assume the ratings follow a normal distribution with a mean of 50 and a standard deviation of 15

- (i) Determine the lowest rating you would give to an “exemplary” employee if you follow the rating guidelines
- (ii) Determine the highest rating you would give to an “competent” employee if you follow the rating guidelines
- (iii) Develop the lowest and highest score in each category